UNIVERSITY HR ANALYTICS

Project Report

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Tools: Microsoft Excel • Power Query • Power BI

1. Project Overview

This project explores a university human-resources dataset supplied by a coach for training purposes. The dataset may be simulated; its real-world status has not been independently confirmed. Power Query was used to prepare the data, and Power BI was used to create an interactive dashboard summarising staff, departments, salary totals, qualifications and recorded publications.

2. Project Objectives

- Summarise staff numbers and departments represented in the dataset.
- Explore salary totals across departments and qualification groups.
- Compare recorded publication counts across qualification groups and departments.
- Provide interactive filters to explore salary and publication records.
- Document data-preparation decisions and communicate limitations clearly.

3. Dataset Description

The prepared UniversityHRAnalytics table contains 2,000 rows and 16 columns. Fields support analysis of staff, departments, salary, qualifications, area of specialisation and publication records. The supplied dataset did not specify a salary currency or period. Therefore, the dashboard reports Total Salary as

685.24M without assigning a currency symbol or assuming whether the amount is monthly, annual or for another period.

4. Data Cleaning and Preparation

Data preparation was carried out in Power Query. The applied steps were documented in the project cleaning log. The process included:

- Promoting headers and assigning appropriate data types.
- Trimming and cleaning text values to reduce unwanted spaces and inconsistent text.
- Replacing selected errors and inconsistent values where required.
- Replacing missing Number of Publications values with 0.
- Replacing missing Area of Specialization values with “unknown”.

The cleaning review identified 880 rows with missing values in Number of Publications at the checked stage; Area of Specialization also contained missing values. Replacing missing publication values with zero is an assumption: it treats a missing entry as zero recorded publications. It does not prove the staff member had no publications. Original records should be checked before any real administrative decisions.

5. Dashboard and Key Performance Indicators

The Power BI dashboard displays these headline metrics:

- Total Staff: 2,000
- Number of Departments: 16
- Total Salary: 685.24M (currency and salary period not specified)
- Recorded Publications: 69,739

Visualisations included:

- Top 10 Departments by Total Salary.
- Salary Distribution by Qualification.

- Recorded Publications by Qualification.
- Top 10 Departments by Recorded Publications.

Dropdown slicers for Salary and Number of Publications allow users to filter the report and explore the visuals under different selections.

6. Observations and Interpretation

The dashboard brings workforce and recorded-publication information together in one report. Department-level salary totals can be compared to identify groups that may warrant further review. Qualification charts allow users to explore how salary and recorded publication totals vary across qualification categories. The department publication chart provides a comparative view of recorded output across departments.

These are descriptive comparisons only. The dashboard does not establish why salary totals differ, whether differences are fair, or whether qualification causes higher salary or publication output. Such conclusions require additional context, validation and appropriate analysis.

7. Recommendations

- Confirm salary currency and period with the dataset provider before adding a currency symbol or interpreting the total.
- Review missing publication and specialisation records against the original source where possible.
- Keep missing values distinct from true zero values in future versions if the source data permits it.
- Use consistent department and qualification labels when updating the dataset.
- When comparing departments, consider staff counts and relevant context alongside salary and publication totals.
- Refresh and validate the report whenever updated source data becomes available.

8. Limitations

- The dataset was supplied for training and may be simulated; this has not been independently confirmed.
- The currency and time period for salary values were not specified.
- Missing publication values were replaced with zero, which may affect totals and comparisons.
- The report describes patterns in the available data and does not prove causal relationships.
- Findings should not be treated as verified facts about a real university without source validation.

9. Conclusion

This project demonstrates a workflow for preparing data with Power Query and presenting workforce-related metrics in Power BI. The dashboard consolidates staff, department, salary and recorded-publication measures into an interactive view. Documenting assumptions and limitations helps users interpret the results responsibly.

10. Project Files

The project folder contains the Power BI report, dashboard images, raw and cleaned datasets, presentation materials and documented data-cleaning steps. These supporting files make the analysis easier to review and reproduce.